

Grade 8
Unit 3
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

Use a calculator to complete the table. Find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
1. $\frac{10^5}{10^{12}}$	.000001	10^{-7}
2. $10^6 + 10^4$	101000	Not Possible
3. $10^7 \times 10^2$	1,000,000,000,000	10^9
4. $(10^5)^3$	1,000,000,000,000,000	10^{15}
5. $10^9 - 10^2$	999999900	Not Possible

6. Which mathematical statements could be rewritten using a power of 10? Explain your reasoning.

$$\frac{10^5}{10^{12}}, 10^7 \times 10^2, (10^5)^3, \frac{1}{10^8}$$

Only numbers that have a 1 followed by all 0's, or is preceded by only 0's, can be written as a power of 10.

7. Which mathematical statements could not be rewritten using a power of 10? Explain your reasoning.

$$10^6 + 10^4$$

$$10^9 - 10^2$$

Only numbers that have a 1 followed by all 0's, or is preceded by only 0's, can be written as a power of 10.

8. Determine which of the mathematical statements align with a law of exponents. Write the law in the table. If no law applies, then write "none".

Mathematical Statement	Laws of Exponents
$\frac{10^5}{10^{12}}$	Quotient of Powers Law
$10^6 + 10^4$	None
$10^7 \times 10^2$	Product of Powers Law
$(10^5)^3$	Power of a Power Law
$\frac{1}{10^8}$	Quotient of Powers / Zero Exponent Law
$10^9 - 10^2$	None

Numbers like a quadrillion, 1,000,000,000,000,000 and a septillion, 1,000,000,000,000,000,000,000,000 are often used by scientists. When using such large numbers, scientists have developed a shortened method of writing these numbers called scientific notation. Scientific notation uses the powers of 10 to represent these numbers.

9. Write 1,000,000,000,000,000 using a power of 10.

$$10^{14}$$

10. Write 1,000,000,000,000,000,000,000,000 using a power of 10.

$$10^{23}$$